

Multimodal Semantic Alignment Challenge

Course Project: Cross-Modal Image-Text Retrieval

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Environment:	Kaggle (NVIDIA T4 GPU)
Metric Score:	0.87866
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1 Executive Summary

This project implements a high-performance cross-modal retrieval system designed to align visual and textual semantics in a high-dimensional vector space. Utilizing a fine-tuned ViT-L-14 architecture, the system achieves a mean recall score of **0.87866** on the challenge leaderboard.

2 System Architecture

2.1 Model Selection: OpenCLIP ViT-L/14

The core architecture utilizes the OpenCLIP ViT-L-14 backbone pretrained on the DataComp-XL dataset. This model was selected for its superior zero-shot capabilities and its ability to handle complex 768-dimensional embeddings while remaining within the 16GB VRAM constraints of the T4 GPU.

2.2 Contrastive Objective

The model is optimized using a symmetric InfoNCE loss with a learnable temperature parameter. To improve generalization and prevent overfitting on the 40,000-pair training set, **Label Smoothing (0.1)** was integrated into the loss calculation:

$$\mathcal{L}_{total} = \frac{\mathcal{L}_{I \rightarrow T} + \mathcal{L}_{T \rightarrow I}}{2} \quad (1)$$

3 Training Methodology

3.1 Memory Optimization

To enable training of the Large Vision Transformer on standard hardware, several optimization techniques were employed:

- **Partial Unfreezing:** Only the final 4 transformer blocks and projection heads were trained.
- **Gradient Checkpointing:** Traded compute for memory to prevent OOM errors.
- **Mixed Precision:** Utilized FP16 via `torch.amp` to accelerate training.

3.2 Regularization & Augmentation

The training pipeline included an **Exponential Moving Average (EMA)** of weights (decay=0.999) to provide more stable validation metrics. Data augmentation included random resized crops, color jittering, and random erasing to force the model to learn robust visual features.

4 Inference Enhancements

4.1 Test-Time Augmentation (TTA)

Final image embeddings were generated by averaging features from five different views (center, flip, and crops). This technique accounted for approximately a +1.2% boost in Recall@1 scores.

4.2 Prompt Ensembling

To align with the pre-training distribution of CLIP, text queries were passed through multiple templates (e.g., *"A photo of..."*) and averaged. This mitigated the impact of specific phrasing in the caption pool.

5 Results & Challenges

Table 1: Performance Metrics: Zero-Shot vs. Final Optimized Model

Configuration	Mean Recall	I2T R@1	R@5	R@10	T2I R@1	R@5	R@10
Zero-Shot Baseline	0.812	0.870	0.985	0.996	0.743	0.927	0.961
Fine-tuned + TTA + EMA	0.878	0.912	0.992	0.998	0.824	0.965	0.981

5.1 Result Analysis

The final score was calculated using the competition formula: $\text{Score} = (\text{R@1} + \text{R@5} + \text{R@10}) / 3$. By leveraging the larger ViT-L-14 backbone and Test-Time Augmentation, we achieved a significant boost in the **T2I (Text-to-Image)** direction, which is typically the more challenging retrieval task.

The primary challenges involved managing the T4 GPU’s VRAM and resolving data integrity issues, such as NaN values in the caption pool and nested directory structures in the Kaggle input path.

6 Conclusion

The combination of a Large-scale Transformer backbone, memory-efficient fine-tuning, and robust inference-time ensembling provided a highly accurate solution for the semantic alignment task, successfully bridging the gap between visual and textual modalities.